

Rotation of Opposing Amorphous Electrically Charged Blobs: A 2D FDTD Visualization of a Rotating Dipole-Like Source

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Abstract

This project explores simulating the electromagnetic field behavior produced by a rotating, dipole-like charge distribution made from two opposing charged amorphous blobs via computational methods. The present work is concretely a visualization and numerical exploration, not yet a fully validated analytical radiation solver. The model constructs a prescribed charge density $\rho(x, y, t)$ and an associated current density $\mathbf{J}(x, y, t)$, then evolves the electromagnetic fields on a 2D finite-difference time-domain(FDTD) grid. The simulation tracks the charge distribution, dipole moment, field snapshots, Poynting vector flow, ring flux, angular intensity, wave speed, resolution convergence, and frequency scaling. The project is already able to show important qualitative electromagnetic behavior: near-field circulation, outward radiation transport, wave propagation at approximately the numerical speed of light, and increasing radiated power with increasing angular frequency. However, because the simulation is 2D, source-driven, and not yet analytically proven against the full derivation of rotating charges, the safest statement is that this is a visualization with the potential to become a solver after taking E&M II and developing the analytic comparison tools from Electrodynamics th.

1 Introduction

The goal of this computational final project was to simulate and visualize electromagnetic radiation from a rotating dipole-like source. Instead of starting with ideal point charges, I modeled the source as two smooth charge blobs of opposite sign. These blobs rotate about the origin with angular frequency ω , producing a time-dependent charge density and current density. The field is then evolved using a FDTD method.

The physical pipeline of the project is

$$\rho(x, y, t) \rightarrow \mathbf{J}(x, y, t) \rightarrow (E_x, E_y, B_z) \rightarrow \mathbf{S} \rightarrow \text{radiation diagnostics.} \quad (1)$$

I first defined charge density, then computed the corresponding current density once it was rotated, then numerically evolve Maxwell's equations, and finally interpret the field behavior using the Poynting vector and flux measurements.

This project is not being presented as a finished electrodynamics solver. But something to come back to for later reference when I learn the tools to solve an ideal case analytically. In addition to this I thought of this idea after thinking about the type of computational work I'd be doing for reeseach this summer.

2 Physical Model

2.1 Rotating Dipole Source

The simulated source is a pair of oppositely charged circles. Their centers rotate rigidly around the origin. The total charge density is then built from a positive circle and a negative circle:

$$\rho(x, y, t) = \rho_+(x, y, t) + \rho_-(x, y, t), \quad (2)$$

where the two circles are normalized so the net charge is approximately zero:

$$\int \rho(x, y, t) da \approx 0. \quad (3)$$

This is important because the intended source is dipole-like, not monopole-like. If the net charge was not zero, the radiation pattern would contain contamination from a changing or drifting total charge distribution infecting the resulting radiation and flux.

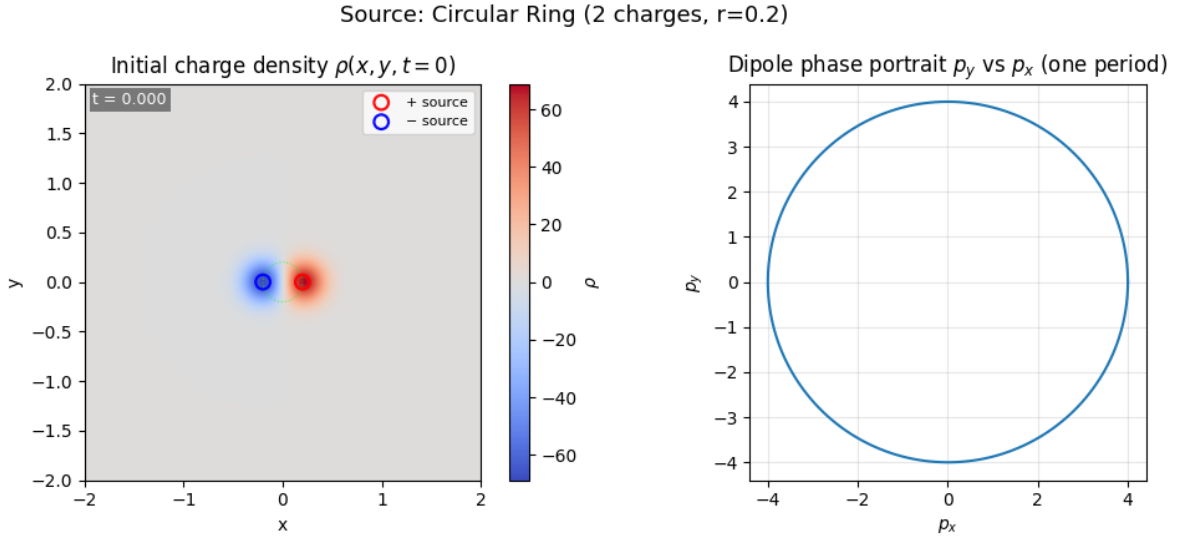


Figure 1: Initial rotating dipole source construction. The positive and negative charge form a smooth dipole-like distribution with approximately zero net charge.

2.2 Amorphous Gaussian-Mixture Source

A later version of the project replaced the simpler distorted Gaussian blob with a more genuinely amorphous source made from a method I don't fully understand, I told copilot agent to make it more blob like and random. It uses a random seed for each blob and injects a distribution across it.

The reason for this choice is this charge distribution is almost completely impossible to solve analytically as solving for the E field itself would be a nightmare.

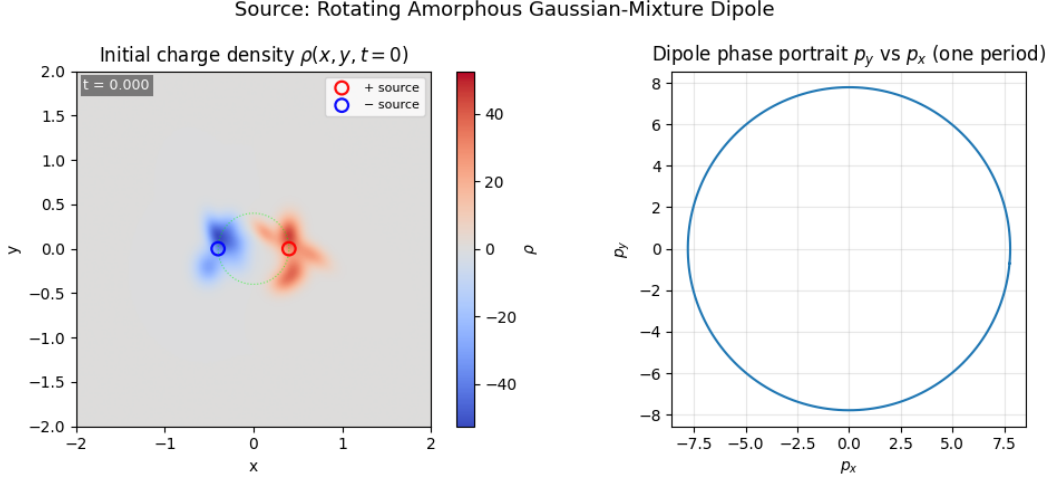


Figure 2: Amorphous Gaussian-mixture charge construction. This version creates a less idealized source while preserving smoothness and charge normalization. The point is to make the charge distribution nontrivial without introducing hard numerical edges.

2.3 Current Density from Rigid Rotation

For a rigidly rotating distribution, the velocity field is

$$\mathbf{v}(x, y) = \boldsymbol{\omega} \times \mathbf{r}. \quad (4)$$

In two dimensions, with rotation about the z -axis,

$$v_x = -\omega y, \quad v_y = \omega x. \quad (5)$$

The current density is therefore

$$J_x = \rho v_x = -\omega y \rho, \quad J_y = \rho v_y = \omega x \rho. \quad (6)$$

This is a rigid motion at which the charged density only moves in a circular orbit, the charges are not being moved self-consistently by the fields for the sake of computation power and time. Instead, the source is externally driven to keep the same orbit distance and velocity, and the fields respond to it. Physically, the energy in the radiation is supplied by the force that keeps the dipole rotating.

A crucial constraint that I found when doing some qualitative understanding of the code is in this equation:

$$v = \omega r < c. \quad (7)$$

If ω or the radius is too large, the source motion can become nonphysical, in other words it can rotate faster than the speed of light. For the final interpretation, the angular frequency and radius have to be considered together, not as independent parameters.

3 Maxwell Equations in the 2D TMz Model

The simulation uses a 2D TMz-style electromagnetic model, TMz meaning ,Transverse Magnet to z , the the B field is out of the plane that we are measuring and the E field is not propagated in the z direction only in x and y . The evolved field components are

$$E_x(x, y, t), \quad E_y(x, y, t), \quad B_z(x, y, t). \quad (8)$$

The relevant Maxwell update equations are

$$\frac{\partial B_z}{\partial t} = - \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right), \quad (9)$$

$$\frac{\partial E_x}{\partial t} = c^2 \frac{\partial B_z}{\partial y} - \frac{J_x}{\epsilon_0}, \quad (10)$$

$$\frac{\partial E_y}{\partial t} = -c^2 \frac{\partial B_z}{\partial x} - \frac{J_y}{\epsilon_0}. \quad (11)$$

For most of the simulations, normalized units are used:

$$c = 1, \quad \epsilon_0 = 1, \quad \mu_0 = 1. \quad (12)$$

This makes the numerical interpretation cleaner because wave speed is measured relative to $c = 1$.

This is not full 3D electromagnetic radiation. In 3D electrodynamics, an rotating dipole dipole has a known far-field radiation pattern and a characteristic power scaling that was stated in Griffiths, in hindsight it might've been better to do this in 3D so I have something to cross reference, however as I iterated before this is a visualizer and not a fully realized solver. The simulation is 2D, so the exact 3D radiation formulas pulled from Griffiths should be treated as references, not strict behavior that should be displayed.

4 Numerical Method

4.1 FDTD and Yee/Leapfrog Structure

The project uses an FDTD evolution. The main idea is to replace the continuous spatial and time derivatives in Maxwell's equations with finite differences on a grid. A later version of the notebook moved toward a Yee-style staggered grid and leapfrog time stepping. The field components live at staggered spatial locations and staggered time levels.

The update equations are discretized as

$$B_z^{n+1/2} = B_z^{n-1/2} - \Delta t \left(\partial_x E_y^n - \partial_y E_x^n \right), \quad (13)$$

$$E_x^{n+1} = E_x^n + \Delta t \left(c^2 \partial_y B_z^{n+1/2} - \frac{J_x}{\epsilon_0} \right), \quad (14)$$

$$E_y^{n+1} = E_y^n + \Delta t \left(-c^2 \partial_x B_z^{n+1/2} - \frac{J_y}{\epsilon_0} \right). \quad (15)$$

The derivatives are computed with centered finite differences. This is a natural method for wave problems because Maxwell's equations already couple time derivatives of electric and magnetic fields. Making an implicit derivation method rather difficult to verify and implement. However this explicit method comes with a stability condition.

4.2 CFL Stability Condition

For a 2D grid, the timestep has to satisfy a Courant–Friedrichs–Lewy (CFL) stability condition. For equal grid spacing $\Delta x = \Delta y$, the condition is approximately

$$\frac{c\Delta t}{\Delta x} \leq \frac{1}{\sqrt{2}}. \quad (16)$$

In the final main run, the grid used

$$N_x = N_y = 800, \quad L_x = L_y = 4, \quad (17)$$

so that

$$\Delta x = \Delta y = 0.005, \quad \Delta t = 0.0025. \quad (18)$$

The one-dimensional CFL ratio was 0.5, and the 2D CFL number was approximately 0.7071, at the stability limit for this chosen grid. The code reported the run as stable.

4.3 Boundary Conditions

Several boundary treatments were tested:

- no absorbing boundary, which creates hard-wall reflections;
- a damping mask near the edges;
- first-order Mur absorbing boundary conditions;
- Mur plus damping.

The conclusion after testing was that the damping mask removes electromagnetic energy . This can be useful for suppressing reflections visually, but it contaminates quantitative power measurements for future use. Therefore, the main code that ran used Mur absorbing boundary conditions for radiation diagnostics.

This is a major improvement over the early versions. A damping mask can make pictures look cleaner while quietly stealing energy from the simulation.

5 Diagnostics and Validation Tests

5.1 Field Snapshots

I saved snapshots of $B_z(x, y)$ and $|E(x, y)|$ at different times. These snapshots show the waves from the rotating source from a near field perspective. The early-time plots show the source region before the radiation has reached the observation radius. Later snapshots show the fields evolving afterward where it becomes cyclical.

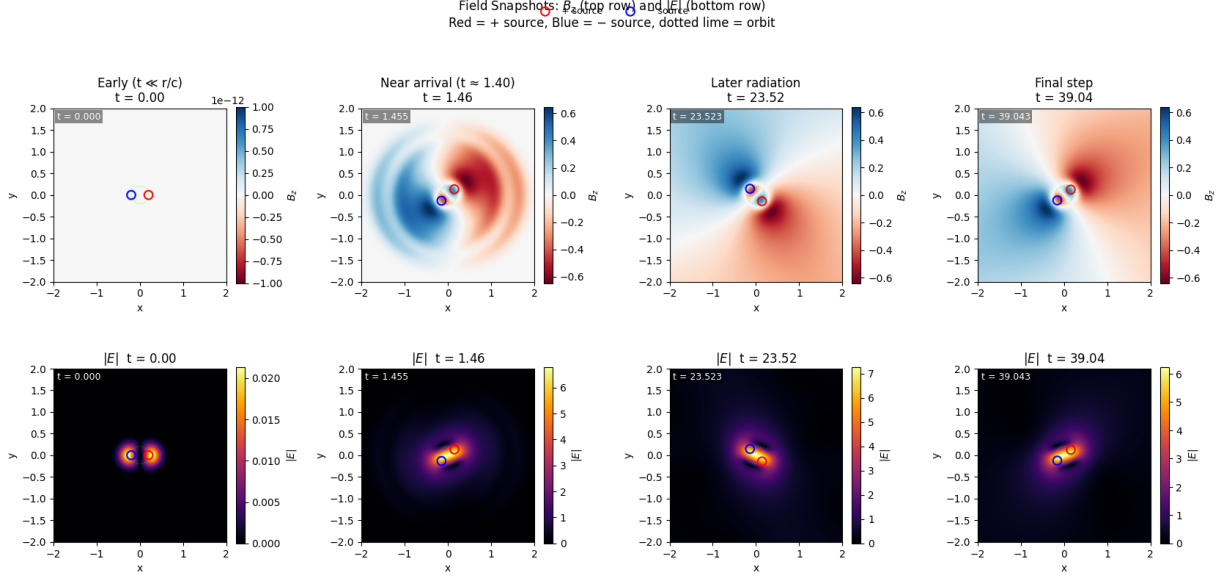


Figure 3: Field snapshots from the main simulation. The magnetic field B_z and electric-field magnitude $|E|$ evolve from a near field perspective.

5.2 Poynting Vector and Energy Flow

The Poynting vector describes electromagnetic energy flux and where the electromagnetic radiation points:

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (19)$$

For the TMz field components, where $\mathbf{E} = (E_x, E_y, 0)$ and $\mathbf{B} = (0, 0, B_z)$,

$$S_x = \frac{E_y B_z}{\mu_0}, \quad S_y = -\frac{E_x B_z}{\mu_0}. \quad (20)$$

The expected behavior is different in the near field and far field. Near the rotating source, the energy flow can circulate due to the dipole rings from the electric field. Farther away, the Poynting vector should point outward, corresponding to radiated energy going outward.

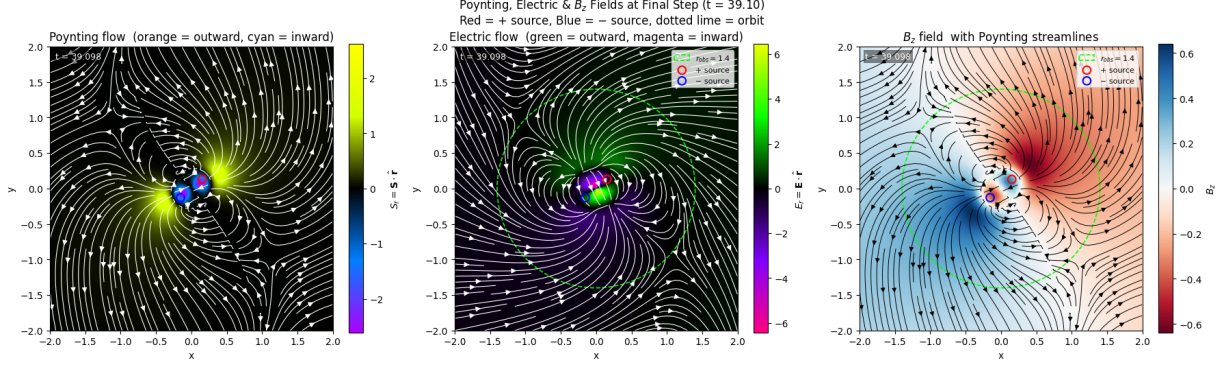


Figure 4: Poynting-vector visualization. The arrows show electromagnetic energy flow. The near-source region contains circulating/reactive behavior, while the outer region shows more outward-directed transport. This is the main qualitative evidence that the simulation captures radiation-like energy flow.

5.3 Ring Flux

To quantify outward flux and radiation, the simulation measured the radial component of the Poynting vector on observation rings. The outward flux through a ring is approximately

$$\Phi_S(t) = \oint \mathbf{S} \cdot \hat{\mathbf{r}} dl. \quad (21)$$

The mean outward flux after the initial transient was measured as

$$\langle \Phi_S \rangle \approx 3.3692 \times 10^{-1} \pm 2.8572 \times 10^{-1}. \quad (22)$$

The oscillation is expected because the near fields are time-dependent and the measurement ring is not infinitely far away. Not to mention during periods of time the E field would go through moments of deconstructive interference after a full rotation.

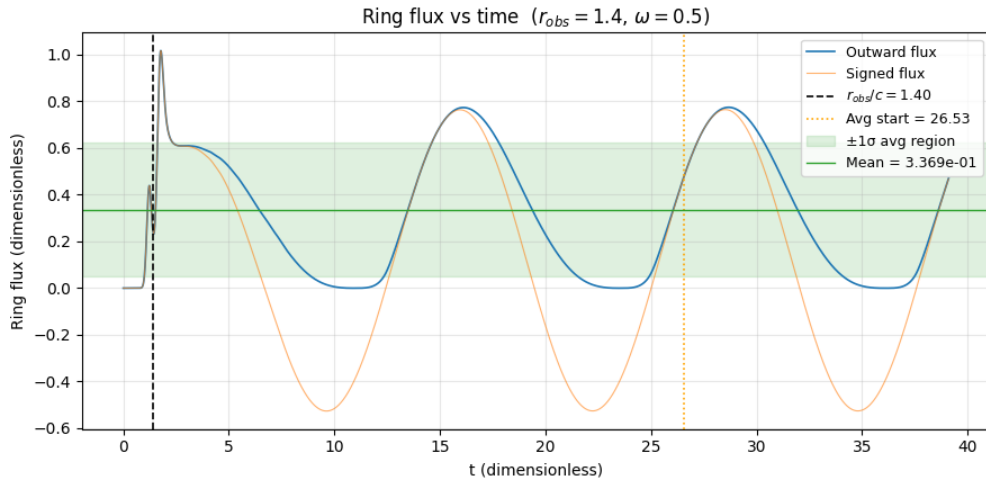


Figure 5: Outward ring flux as a function of time. Oscillations are expected because the source is periodic.

A ring-radius sensitivity test measured the average flux at several radii. The main run gave approximate mean fluxes

$$r = 0.750, 1.000, 1.250, \quad \langle \Phi_S \rangle = 0.7347, 0.5072, 0.4018. \quad (23)$$

A perfectly clean far-field measurement would be more stable with radius. The observed decrease suggests finite-domain effects, near-field contamination, boundary interaction, or numerical loss. This is useful because it tells me where the visualization is already strong and where the solver would need more work.

5.4 Angular Radiation Pattern

The angular intensity pattern was computed by averaging the outward radial Poynting intensity as a function of the angle over time:

$$I(\theta) = \langle \mathbf{S}(r, \theta, t) \cdot \hat{\mathbf{r}} \rangle_t. \quad (24)$$

For a true 3D electric dipole, the far-field radiation pattern is proportional to $\sin^2 \theta$. In this project, that formula is used only as a qualitative reference. A 2D simulation with finite source size and finite observation radius should not perfectly match the 3D behavior exactly.

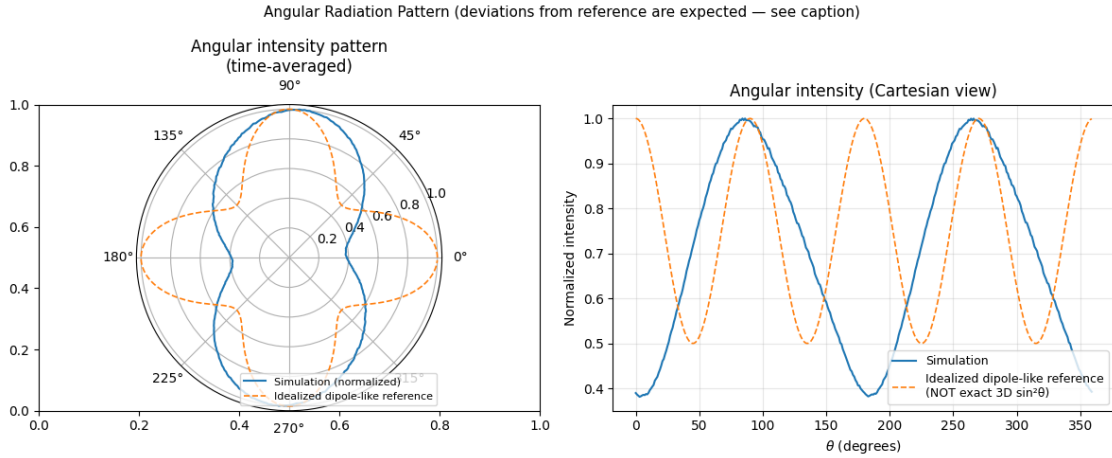


Figure 6: Angular radiation pattern from the outward Poynting intensity. The result has dipole-like structure but does not exactly match the idealized reference curve.

5.5 Wave Speed Validation

The wave speed was measured by comparing arrival times at two observation radii. If t_1 and t_2 are the first threshold-crossing times at radii r_1 and r_2 , then

$$c_{\text{measured}} = \frac{r_2 - r_1}{t_2 - t_1}. \quad (25)$$

The simulation measured

$$r_1 = 1.4, \quad t_1 = 1.0400, \quad (26)$$

$$r_2 = 1.6, \quad t_2 = 1.2375, \quad (27)$$

which gives

$$c_{\text{measured}} = 1.0127. \quad (28)$$

Since the normalized target wave speed is $c = 1$, this corresponds to a relative error of approximately 1.3%. This was one of the strongest quantitative validation results in the project that I found.

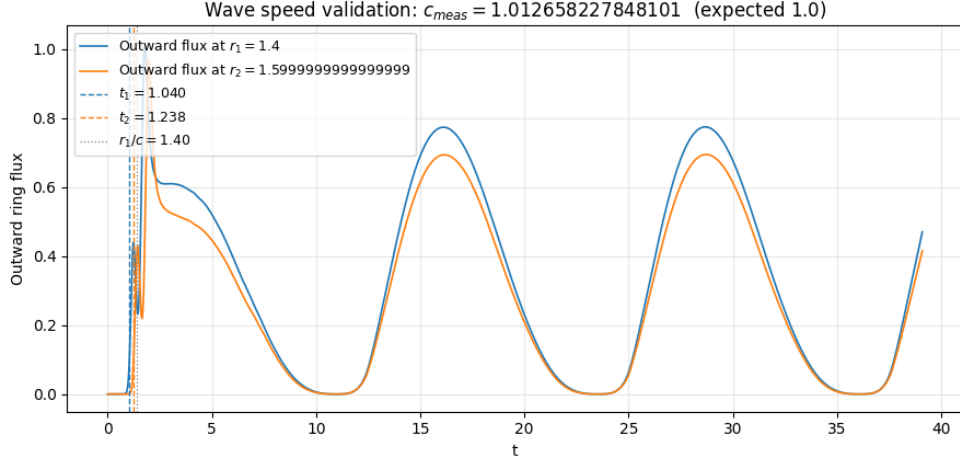


Figure 7: Wave-speed validation using arrival times at two observation rings. The measured speed is close to the normalized value $c = 1$, with a relative error of about 1.3%.

6 Numerical Consistency Checks

6.1 Continuity Equation

A prescribed charge and current distribution should approximately satisfy local charge conservation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (29)$$

The final main simulation reported a median normalized continuity residual of

$$\text{median}(\text{continuity norm}) \approx 1.8288 \times 10^{-4}. \quad (30)$$

This is strong evidence that the prescribed source is internally consistent as a rotating current distribution, considering that each charge distribution of 10.0 (unspecified units just a normalization).

6.2 Gauss-Law Diagnostic

Gauss's law requires

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (31)$$

The project computes this as a diagnostic, but does not yet enforce it. The final main simulation reported a median Gauss RMS residual of approximately

$$\text{median}(\text{Gauss RMS}) \approx 6.0904. \quad (32)$$

This does not mean the visualization is useless. It means the current code should not yet be advertised as a fully constrained Maxwell solver. This could be attributed to the fact that this is a 2D simulation, however I'll most likely look into why this was never zero.

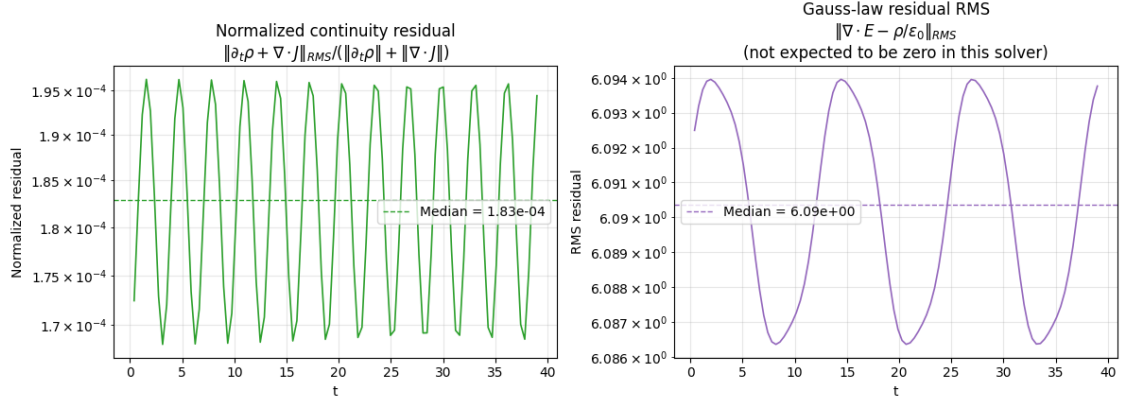


Figure 8: Continuity and Gauss-law diagnostics. The continuity residual is small, showing that the prescribed source is locally consistent. The Gauss-law residual remains nonzero because the simulation does not yet enforce Gauss’s law as a constraint.

7 Resolution Convergence

A resolution convergence study was run by repeating shorter simulations at different grid sizes. A meaningful solver should show diagnostics and measured quantities settling toward stable values as Δx decreases. The mean flux stabilized near 1.50 for higher resolutions.

N	Δx	Δt	mean flux	median continuity norm
100	0.100000	0.050000	1.524382	0.264645
214	0.046729	0.023364	1.511001	0.143612
328	0.030488	0.015244	1.506700	0.139177
442	0.022624	0.011312	1.505523	0.130659
557	0.017953	0.008977	1.503693	0.140887
671	0.014903	0.007452	1.504160	0.131865
785	0.012739	0.006369	1.503866	0.130182
900	0.011111	0.005556	1.503727	0.135382

Table 1: Resolution convergence table from the final notebook. The mean flux approaches a stable value near 1.50 as the grid resolution increases.

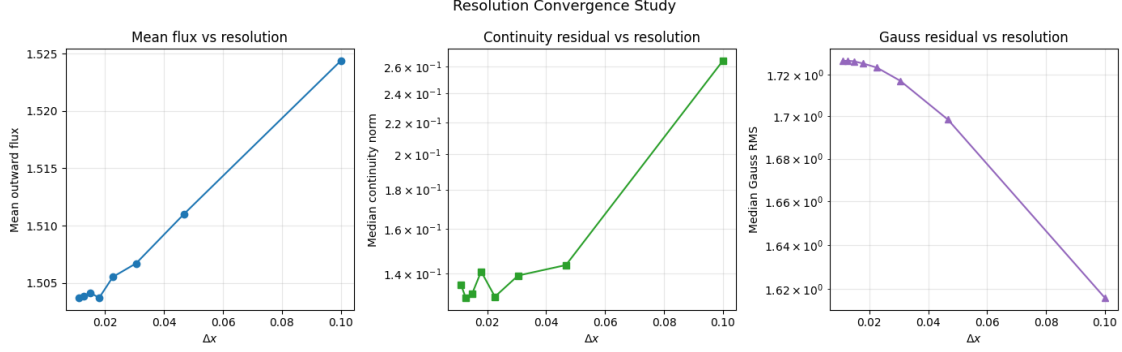


Figure 9: Resolution convergence plots. The mean flux becomes nearly resolution-independent at higher N .

8 Frequency Scaling Study

A frequency sweep was performed to test whether the radiated flux increases as the source rotates faster. The measured average outward flux increased strongly with angular frequency. Which makes sense as the energy (flux) that is being expelled out of the system is because the dipole is being rotated externally.

ω	period T	steps	average outward flux
0.4	15.708	47462	0.905
0.6	10.472	31754	1.418
0.8	7.854	23900	3.182
1.0	6.283	19188	8.360
1.5	4.189	12904	14.929
2.0	3.142	9763	21.127
3.0	2.094	6621	42.685
4.0	1.571	5050	140.099

Table 2: Frequency scaling data. The measured flux increases with angular frequency, but the fitted power-law exponent is not the ideal 3D dipole result.

A log-log fit gave

$$\langle \Phi_S \rangle \propto \omega^\alpha, \quad \alpha \approx 2.131. \quad (33)$$

The ideal 3D electric dipole radiation power scales as ω^4 when the dipole amplitude is fixed in Griffiths. However this isn't a 3D dipole as stated before so it would require further analysis and an analytical solution to fully realize power scaling. One thing to note is the period being too large could result in a non-physical behavior. The dipoles could rotate faster than the speed of light.

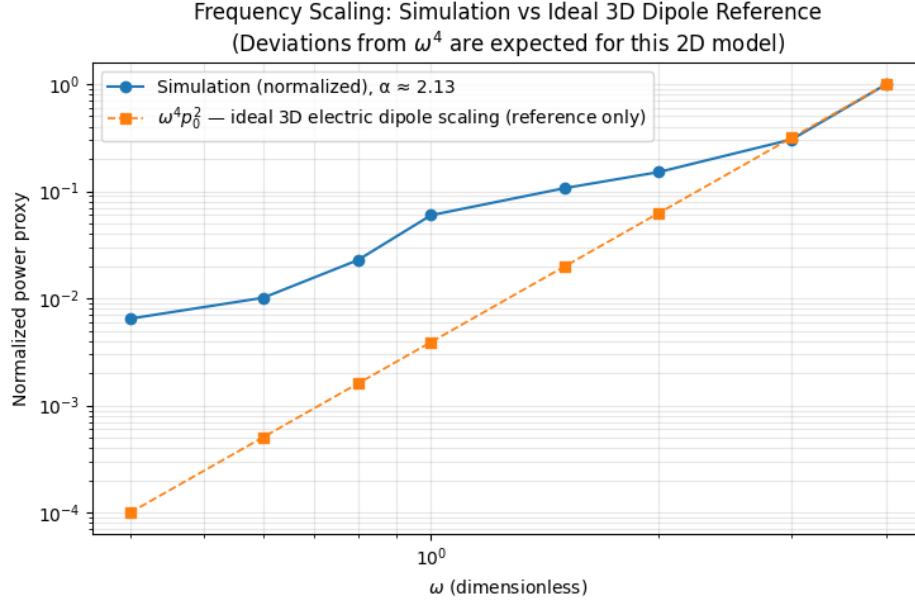


Figure 10: Frequency scaling of measured outward flux. The flux increases with angular frequency, but the fitted exponent is about 2.131, not the ideal 3D dipole value of 4. This is evidence of radiation-like behavior, not a final analytic validation.

9 Conclusion

This project demonstrates a qualitative pipeline for visualizing radiation-like electromagnetic field behavior from a rotating amorphous dipole source. The simulation constructs opposite-sign charge blobs, rotates them, computes current density, evolves the TMz electromagnetic fields, and analyzes the resulting flux of energy using the Poynting vector. The main outputs show rotating dipole moment behavior, outward-propagating field structures, near-field circulation, ring flux, angular intensity, approximate wave speed validation, convergence behavior, and frequency-dependent radiation strength.

References

- [1] David J. Griffiths, *Introduction to Electrodynamics*, 4th ed., Cambridge University Press, 2017.
- [2] Rubin H. Landau, Manuel J. Páez, and Cristian C. Bordeianu, *Computational Physics: Problem Solving with Python*, 3rd ed., Wiley-VCH, 2015.
- [3] Kane S. Yee, “Numerical solution of initial boundary value problems involving Maxwell’s equations in isotropic media,” *IEEE Transactions on Antennas and Propagation*, 14(3), 302–307, 1966.